

Autodesk® Scaleform®

ActionScript 2 XML

Scaleform 4.3 ActionScript 2 XML

: Prasad Silva
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Autodesk® Scaleform® 4.3

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XML

Autodesk Scaleform Corporation

6305 Ivy Lane, Suite 310

Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

(301) 446-3200

(301) 446-3199

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Autodesk® Scaleform® SDK
 PC Scaleform Flash
 Adobe®
 Flash® Studio

Flash AS2 XML API ActionScript 2 XML XML XMLNode
 DOM XML Scaleform 4.1
 Scaleform DOM SAX2 XML
 Scaleform Player Expat Scaleform
 XML Windows®, Linux®, MacOS®, Xbox 360®, PSP® (PlayStation®Portable), PlayStation®2 (PS2),
 PlayStation®3 (PS3™), Nintendo® Wii
 Scaleform XML

1. XML

Scaleform XML

1.1 ActionScript XML

ActionScript XML

Scaleform

XML

FxPlayer.cpp

```
...
#include "XML/XML_Expatri.h"
using namespace Scaleform;
...
Ptr<XML::Parser> pexpatXmlParser = *new XML::ParserExpat;
Ptr<XML::SupportBase> pxmlSupport = *new XML::Support(pexpatXmlParser);
mLoader.SetXMLSupport(pxmlSupport);...
```

1.2 C++

XML

C++
DOM

XML
Scaleform

ActionScript XML

```
...
#include "GFx/XML/XML_DOM.h"
using namespace Scaleform;
...
//          DOM
XML::DOMBuilder domBuilder(Loader.GetXMLSupport());
//          DOM
XML::DOMBuilder domBuilder2(Loader.GetXMLSupport(), true);
...
//  xml      DOM
Ptr<XML::Document> pdoc = domBuilder.ParseFile("inputfile.xml",
                                              mLoader.GetFileOpener());

//  xml      DOM
Ptr<XML::ObjectManager> pObjMgr = *new XML::ObjectManager();
Ptr<XML::Document> pdoc2 = domBuilder.ParseFile("inputfile.xml",
                                              mLoader.GetFileOpener(), pObjMgr);
...
```

ActionScript XML


```

        printf("ELEM - '%s' ns:'%s' prefix:''"
               " localname: '%s'",
               pnode->Value.ToCStr(),
               pnode->Namespace.ToCStr(),
               pnode->Value.ToCStr());
    }
    for (XML::Attribute* attr = pnode->FirstAttribute;
         attr != NULL; attr = attr->Next)
    {
        printf(" {%s,%s}", attr->Name.ToCStr(),
               attr->Value.ToCStr());
    }
    printf("\n");
    for (XML::Node* child = pnode->FirstChild; child != NULL;
         child = child->NextSibling)
        PrintDOMTree(child);
    break;
}
case TextNodeType:
{
    printf("TEXT - '%s'\n", pnode->Value.ToCStr());
    break;
}
default:
{
    printf("UNKN\n");
}
}
}
...
Ptr<XML::Document> pdoc = domBuilder.ParseFile("inputfile.xml",
                                                Loader.GetFileOpener());
PrintDOMTree(root);

```

1.4

```

Scaleform                                XML_DOM.cpp                                XML
XML      DOM
//
//
#ifdef SF_BUILD_DEBUG
////

```



```
//          DOM
// #define SF_XML_DOCBUILDER_TRACE
//
//          DOM
// (
//
// #define SF_XML_DOCBUILDER_DOM_TREE_DUMP
//
#endif // #ifdef SF_BUILD_DEBUG
```

2.

Scaleform			XML
GFx::XML::Parser	Scaleform XML		XML_Support.h
XML_Support.h	GFx::XML::ParserHandler	SAX2	

```
namespace Scaleform { namespace GFx { namespace XML {
// XML
//
//      /
//
// XML
class Parser : public RefCountBaseNTS<Parser, Stat_Default_Mem>
{
public:
    virtual ~Parser() {}

    // Parse methods

    virtual bool    ParseFile(const char* pfilename, FileOpenerBase* pfo,
                             ParserHandler* pphandler) = 0;
    virtual bool    ParseString(const char* pdata, UPInt len,
                               ParserHandler* pphandler) = 0;
};
}}} //SF::GFx::XML
```

XML	DOM	GFx::XML::ParserLocator
ParseFile	ParseString	
GFx::XML::ParserHandler		

```
namespace Scaleform { namespace GFx { namespace XML {
// SAX2
// DOM          SAX2
//
//
//
class ParserHandler : public RefCountBase<ParserHandler, StatMV_XML_Mem>
{
public:
    ParserHandler() {}
    virtual ~ParserHandler() {}
}
```

```

//
virtual void      StartDocument() = 0;
virtual void      EndDocument() = 0;

//
virtual void      StartElement(const StringRef& prefix,
                                const StringRef& localname,
                                const ParserAttributes& atts) = 0;
virtual void      EndElement(const StringRef& prefix,
                              const StringRef& localname) = 0;

//
virtual void      PrefixMapping(const StringRef& prefix,
                                const StringRef& uri) = 0;

//
virtual void      Characters(const StringRef& text) = 0;

//
virtual void      IgnorableWhitespace(const XMLStringRef& ws) = 0;

//
virtual void      SkippedEntity(const XMLStringRef& name) = 0;

// Gfx::XML::Parser                                Gfx::XML::ParserHandler
// // //                                           xml
virtual void      SetDocumentLocator(const ParserLocator* plocator) = 0;

//
virtual void      Comment(const StringRef& text) = 0;

//
virtual void      Error(const ParserException& exception) = 0;
virtual void      FatalError(const ParserException& exception) = 0;
virtual void      Warning(const ParserException& exception) = 0;

};
}}} //SF::Gfx::XML

```


3.2 *ActionScript XML.status*

ActionScript 2.0 XML

ActionScript

- 0
- -2 A CDATA
- -3 XML
- -4 DOCTYPE
- -5
- -6 XML
- -7
- -8
- -9
- -10

ActionScript

```
//  
// XML_ERROR_NONE = 0  
// XML_ERROR_INVALID_TOKEN = 0 (ie: XML.parse("http://www.google.com"))  
// XML_ERROR_UNCLOSED_CDATA_SECTION = -2  
// XML_ERROR_UNCLOSED_TOKEN = -3  
// XML_ERROR_INVALID_TOKEN = -4  
// XML_ERROR_INVALID_TOKEN = -5  
// XML_ERROR_UNCLOSED_TOKEN = -8  
// XML_ERROR_NO_ELEMENTS = -9  
// XML_ERROR_TAG_MISMATCH = -10  
//
```

GfX::XML::DOMBuilder

XML

GfX::XML::ParserLocator

3.3 *ActionScript XML.docTypeDecl*

Flash®

ActionScript XML

DOCTYPE

XML.docTypeDecl

DTD

Flash Scaleform

